

Ciudad Universitaria a 4 de Septiembre, 2026

Wellcome Leap
Resistance Networks Program

Re: Support for “Universal properties of plasmid evolution and robust risk forecasting for preemptive pARG intervention”

Dear Members of the Review Committee:

I am writing in support of the proposal submitted by Dr. Philip J. Gerrish to the Wellcome Leap Resistance Networks program.

Phil and I share a scientific lineage extending from Richard Lenski’s laboratory, and I have known him and followed his work for many years. We have maintained a close scientific relationship throughout much of our careers, particularly through our common interests in microbial evolution. He is one of my longstanding and valued scientific colleagues in Mexico.

From my perspective as a microbial evolutionary ecologist, the proposed work addresses an important limitation in current approaches to antibiotic resistance. The spread of resistance plasmids cannot be understood from antibiotic selection alone. It also depends on microbial community structure, host diversity, ecological conditions, population bottlenecks, and the balance between vertical inheritance and horizontal transfer. Phil’s theoretical framework provides a coherent way to examine these interacting processes and to formulate quantitative predictions from them.

Phil has an unusual ability to identify general evolutionary mechanisms while remaining attentive to the biological assumptions required for their empirical application. His proposed combination of mathematical theory, simulation, and prospective validation is consistent with the work I have known him to conduct over many years.



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His extensive scientific experience in Mexico and his longstanding relationships with Mexican researchers and students are additional strengths. He is well positioned to connect international theoretical work with the biological and public-health problems encountered in Mexico and elsewhere.

I strongly support this proposal and Phil's leadership of the project.

Sincerely,

Dra. Valeria Souza Saldívar

Investigadora Titular C

Emerito SNI

Instituto de Ecología

Universidad Nacional Autónoma de México

Honorary international Member Academy of Arts and Sciences USA

Miembro de la National Academy of Sciences